

Review of Last Lecture

1.1 Introduction

1.2 Signal classification and typical signals

1.3 Signal operations

1.4 Singular signals

Message: A symbol to be transmitted in a way agreed in advance by both the sender and receiver, such as language, text, images, data, etc.

Information: Unknown content obtained from the received message.

Signal: A signal is a **variation** of a measurable quantity that carries **information** about the behaviour of a system. Examples: human voice, room temperature controlled by a thermostat system, position, speed, and acceleration of an aircraft, force measured with force sensors in robotic systems, digital photographs, music recording etc.

Electrical signal: The current or voltage that changes corresponding to the message (language, text, image, data), or the charge on the capacitor, the magnetic flux in the inductor, etc.

1.2 Signal Classification and Typical Signals



1.2.1 Classification of Signals

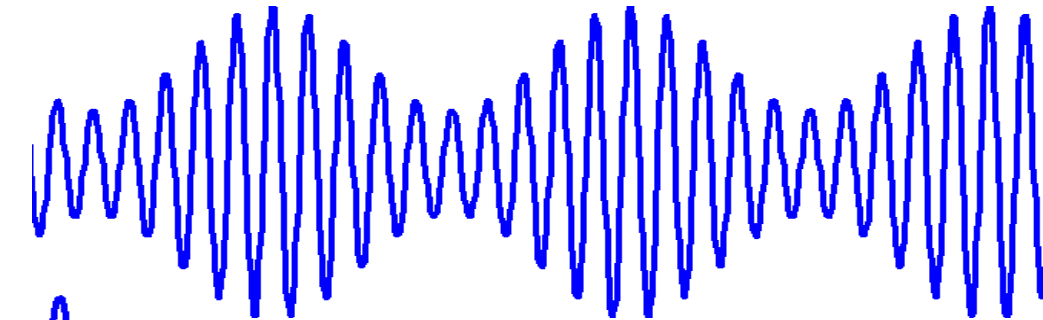
1. Deterministic Signals and Random Signals

Deterministic signal - for a certain moment, the signal has a definite value corresponding to it.

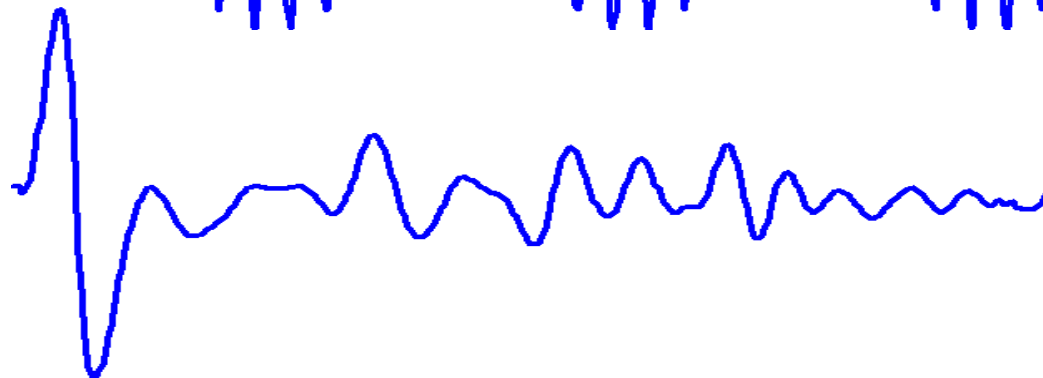
Random signals - unpredictable signals such as noise.

Deterministic signals

$$S(t) = (2 + \cos \omega t) \cos (10 \omega t)$$



Random signal



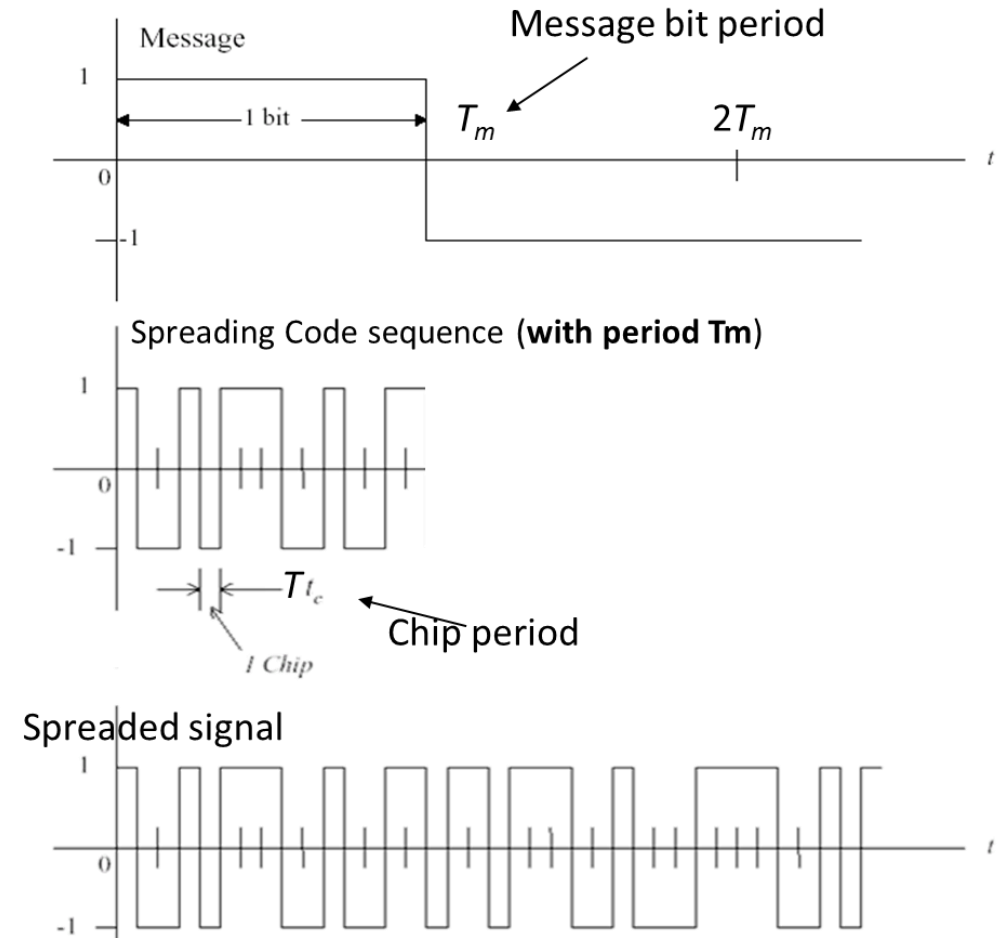
1.2 Signal Classification and Typical Signals



2. Periodic Signals and Non-Periodic Signals

Periodic signal: It starts and starts at a certain time interval, and it is a signal without beginning or end. A signal with a basic period of T - $f(t)=f(t+T)$ for all t .

Non-periodic signal: A signal that does not meet the characteristics of the cycle in time.



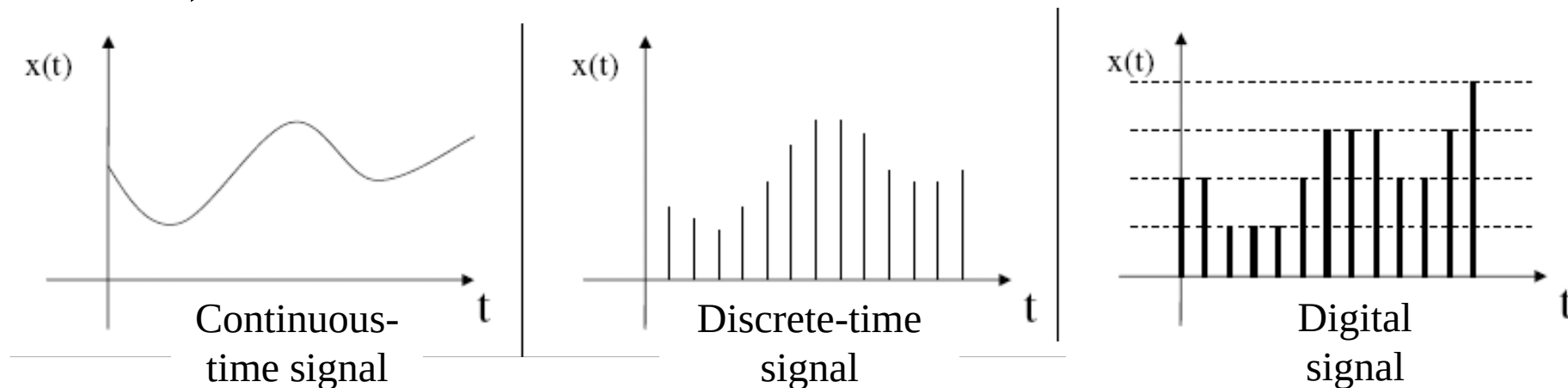
The spreading code of direct-sequence spread spectrum is a periodic signal.

3. Continuous-Time Signals and Discrete-Time Signals

Continuous-time (analogue) signal: Within the time interval under discussion, a definite function value can be given for any time value (except for several discontinuous points).

Discrete-time signal: The signal has values only at discrete time points and is undefined at other times. It is usually obtained via sampling of continuous-time signals.

Digital signal: The signal is discrete in both time and amplitude. **Quantization** is the process of converting a continuous range of analog signal amplitudes (or highly precise discrete values) into a finite set of discrete levels.



$\sin(0.4n)$ is a digital signal. The statement is ()

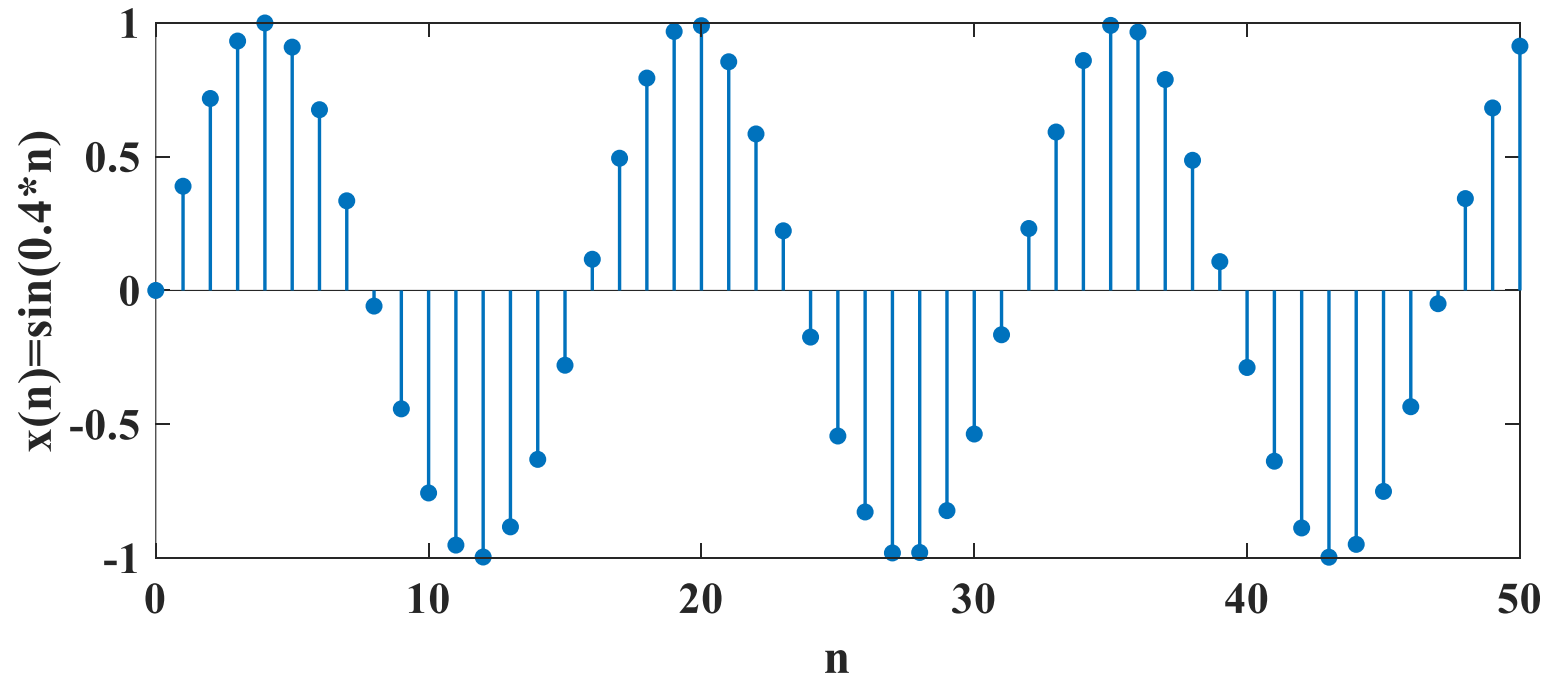
- ☐ A Ture
- ☒ B False

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1.2 Signal Classification and Typical Signals



Sampling $x(t) = \sin(0.4t)$ yields $x(n) = \sin(0.4n)$. If $x(n)$ is a periodic signal, there exists an integer N that satisfies $\sin(0.4n) = \sin[0.4(n + N)]$. On the other hand, one can derive $0.4N = 2k\pi$, i.e., $N = 5k\pi$ (k is an integer), which is an irrational number and cannot be an integer. Therefore, $x(n)$ is a **non-periodic** signal. The amplitude cannot be represented by a finite set of discrete levels. $x(n)$ is a **discrete-time signal**, but **NOT** a **digital signal**.



4. Energy signals and power signals

Energy of real-valued signal $f(t)$: $E = \lim_{T_0 \rightarrow \infty} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} f^2(t) dt$

Average power of signal $f(t)$: $P = \lim_{T_0 \rightarrow \infty} \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} f^2(t) dt$

Energy signal:

$$0 < E < \infty \quad (\text{with finite value}), \quad P = 0$$

Power signal:

$$0 < P < \infty \quad (\text{with finite value}), \quad E = \infty$$

- ① Generally, periodic signals are power signals.
- ② Non-periodic signals with values within a limited interval are energy signals.
- ③ Some non-periodic signals are neither energy signals nor power signals.

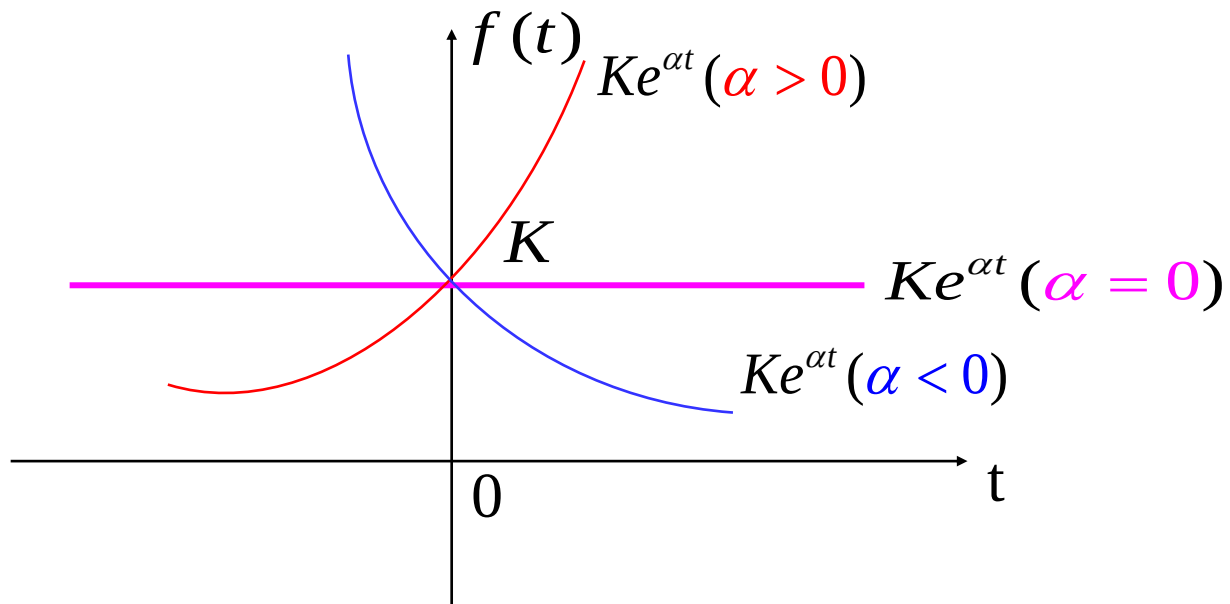
1.2 Signal Classification and Typical Signals



1.2.2 Typical Signals

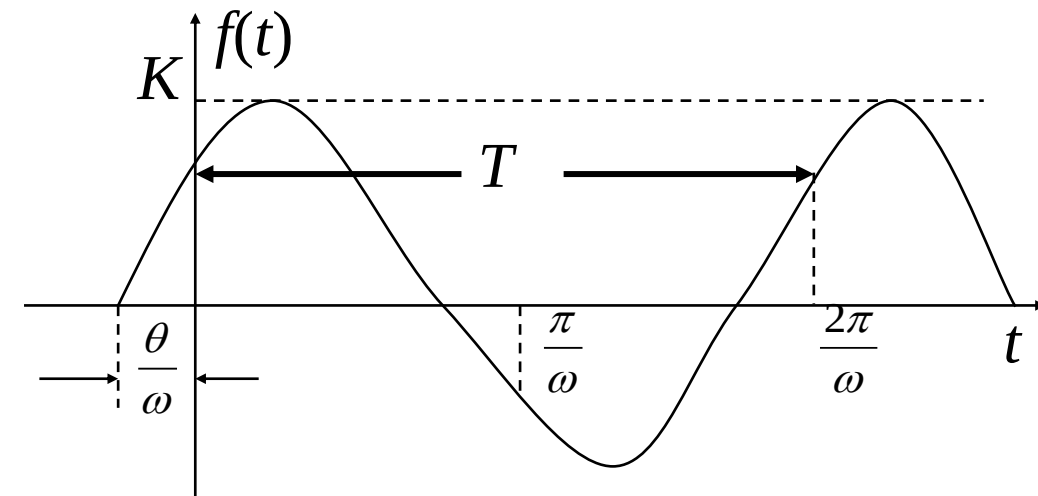
1. Exponential Signal

Expression: $f(t) = Ke^{\alpha t}$



2. Sinusoidal Signal

$$f(t) = K \sin(\omega t + \theta)$$



3. Complex Exponential Signal

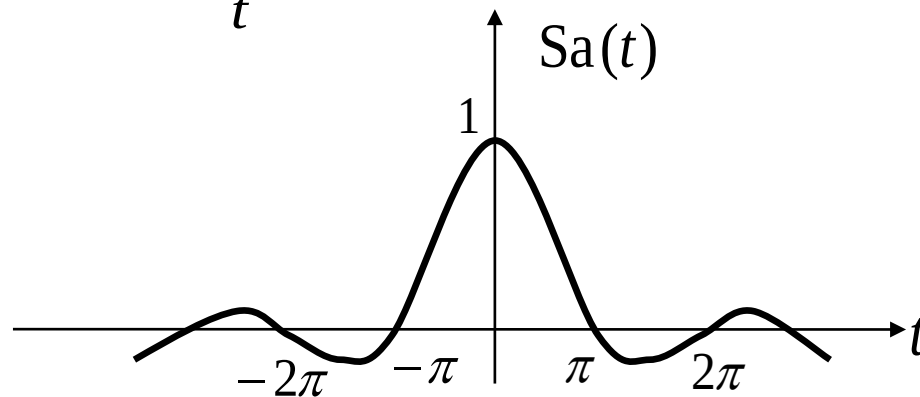
If the exponential factor of an exponential signal is a complex number, it is referred to as a complex exponential signal, expressed as:

$$f(t) = Ke^{st} = Ke^{(\sigma + j\omega)t} = Ke^{\sigma t} \cos \omega t + jKe^{\sigma t} \sin \omega t$$

4. Sampling function $\text{Sa}(t)$ and $\text{Sinc}(t)$

The sampling function is defined as:

$$\text{Sa}(t) = \frac{\sin t}{t}$$



An alternative function is defined as:

$$\text{Sinc}(t) = \frac{\sin(\pi t)}{\pi t}$$

1.3 Signal Operations

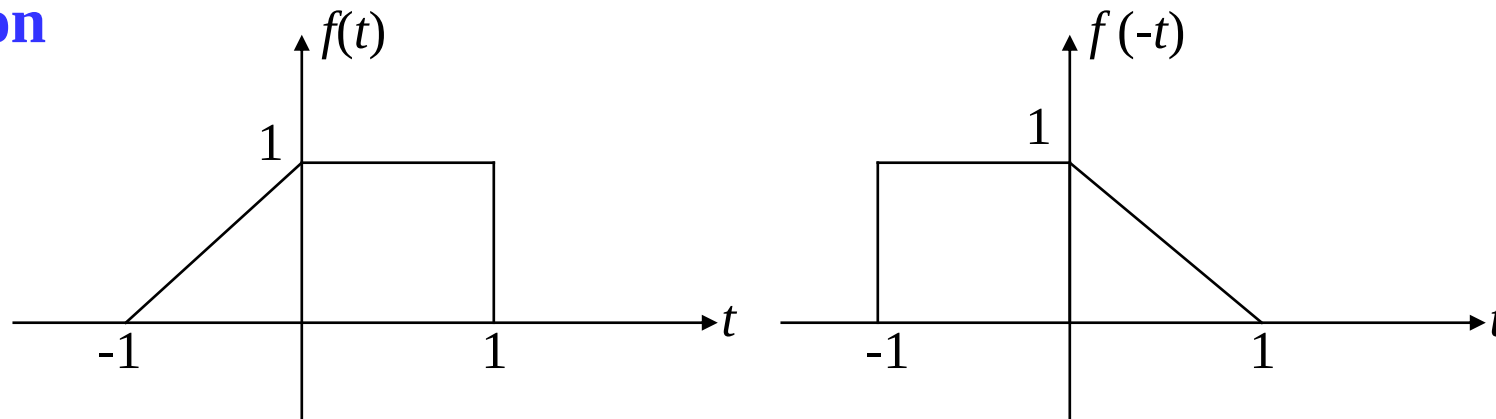


1.3.1 Time Operations-Reverse, Shifting, and Scaling

1. Reverse Operation

$$f(t) \rightarrow f(-t)$$

Reverse against the
 $t = 0$ axis.

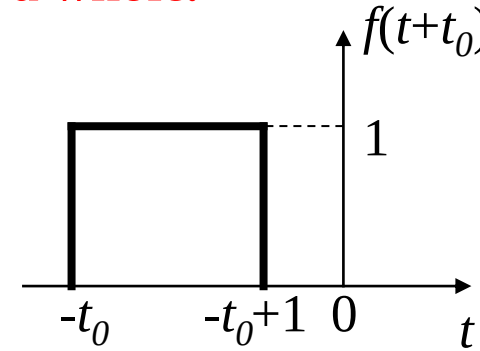
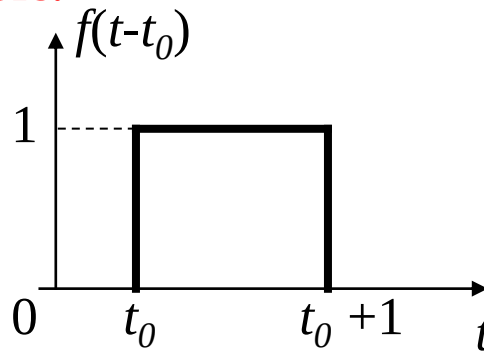
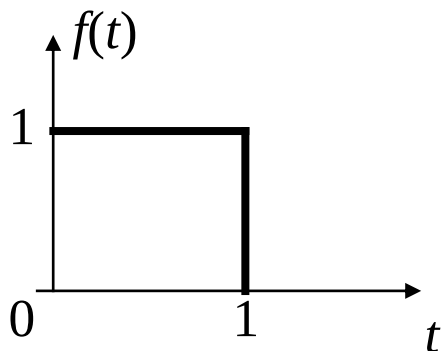


2. Time Shifting Operation

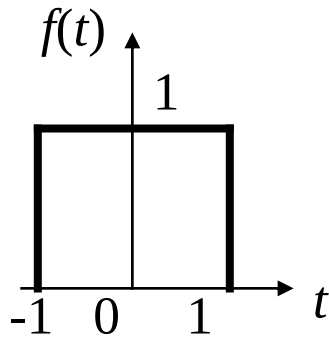
$$f(t) \rightarrow f(t-t_0)$$

$t_0 > 0$ 时, $f(t)$ shifts to the right by t_0 as a whole.

$t_0 < 0$ 时, $f(t)$ shifts to the left by t_0 as a whole.

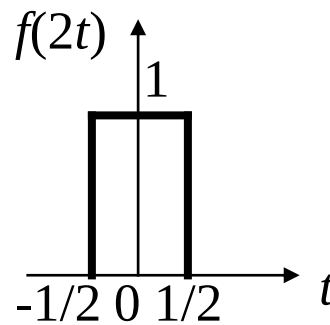


3. Scaling Operation



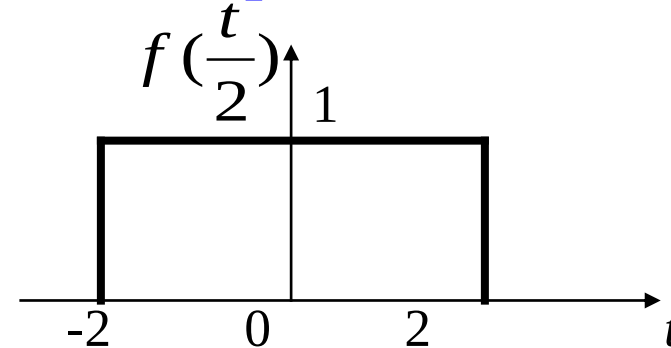
$$f(t) \rightarrow f(2t)$$

Compression



$$f(t) \rightarrow f\left(\frac{t}{2}\right)$$

Expansion



$$f(t) \rightarrow f(at)$$

Compression: If $|a| > 1$, the waveform is compressed along the time axis.

Expansion: If $0 < |a| < 1$, the waveform is expanded along the time axis.

1.3.2 Differentiation and Integration Operations of Signals

1. Differentiation Operation

The derivative of signal $f(t)$ is still a signal, represented by $f'(t)$, which represents the rate of change of the signal with time. The differentiation operation highlights the changing part of the signal.

2. Integration Operation

- The integral of signal $f(t)$, denoted by $\int_{-\infty}^t f(\tau) d\tau$ or $f^{(-1)}(t)$, is still a signal. Its value at any time is equal to the area enclosed by $f(t)$ and the time axis within the interval from $-\infty$ to t .
- The integration operation smooths the abrupt parts of the signal and can reduce the impact of glitches (noise).

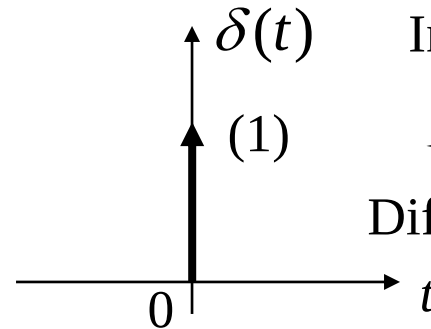
1.4 Singular Signals



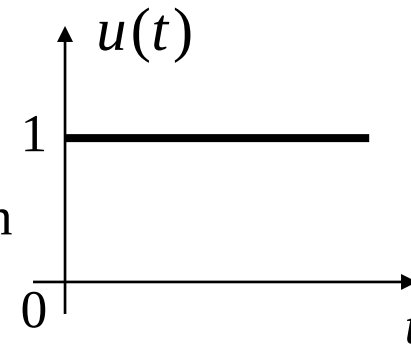
Singular signals: The signals with discontinuities in themselves or their derivatives/integrals.

Relationship between $tu(t)$, $u(t)$, $\delta(t)$, and $\delta'(t)$:

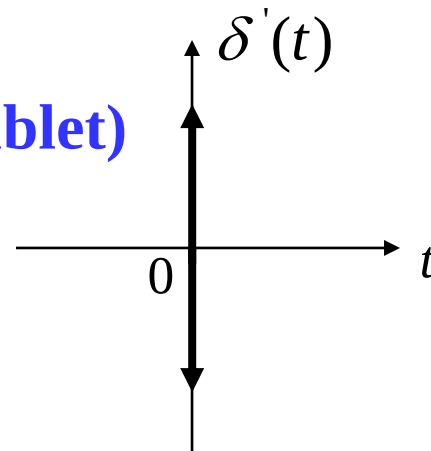
**Unit Impulse Signal
(Dirac Delta Function)**



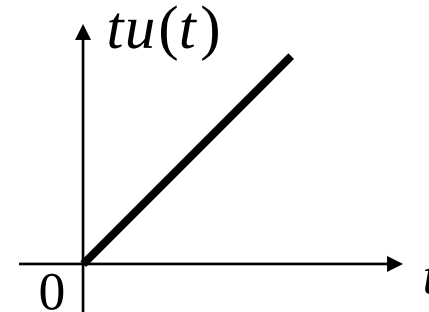
Unit Step Signal



Impulse Derivative (Doublet)



Unit Ramp Signal

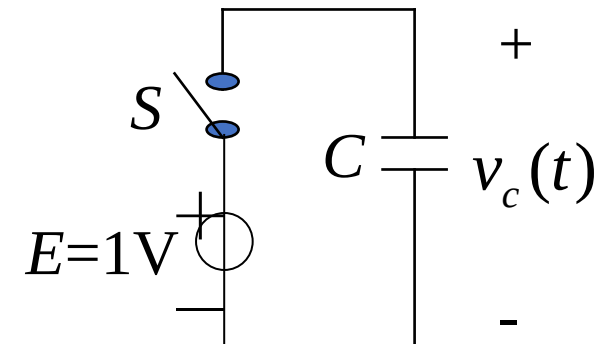


Unit Step Signal

$$u(t) = \begin{cases} 0, & (t < 0) \\ 1, & (t > 0) \end{cases}$$



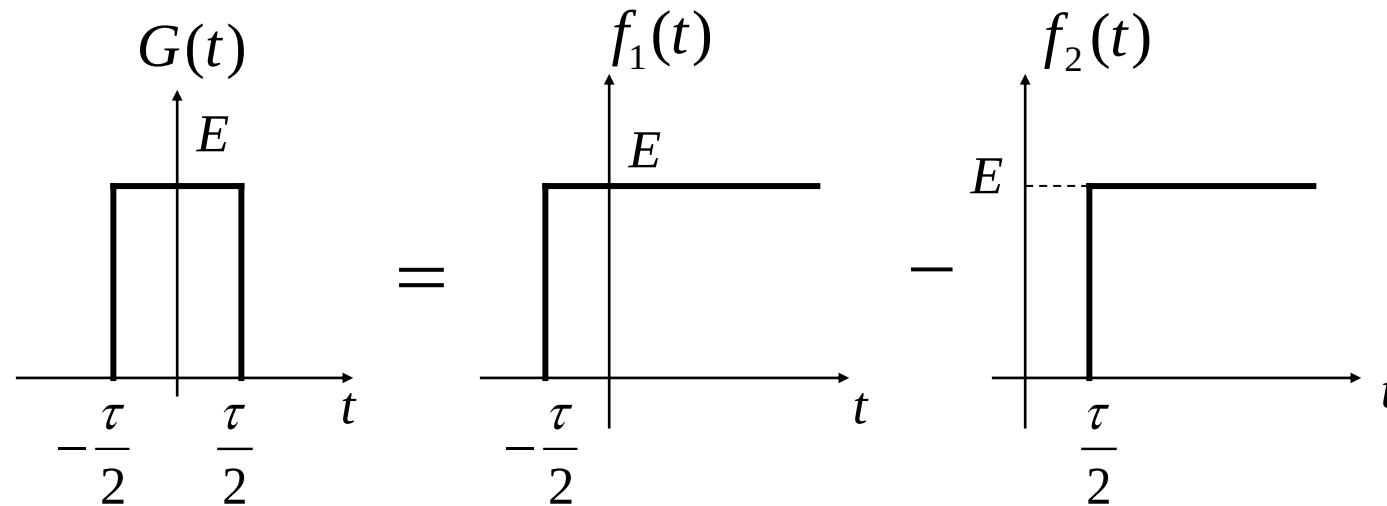
Example: The voltage across the capacitor C when S is closed at $t = 0$ is a unit step signal.



1.4 Singular Signals



Properties of $u(t)$: Unilateral characteristic, i.e., $f(t)u(t) = \begin{cases} 0 & t < 0 \\ f(t) & t > 0 \end{cases}$



Since $f_1(t) = Eu(t + \frac{\tau}{2})$, $f_2(t) = Eu(t - \frac{\tau}{2})$,

the rectangular pulse $G(t)$ can be expressed as:

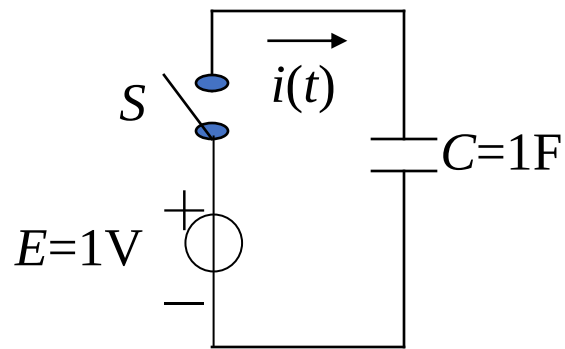
$$G(t) = f_1(t) - f_2(t) = E[u(t + \frac{\tau}{2}) - u(t - \frac{\tau}{2})]$$

1.4 Singular Signals



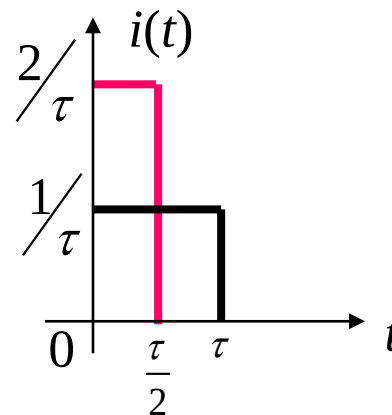
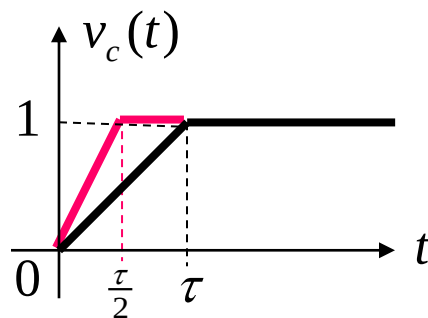
Unit Impulse Signal (Dirac Delta Function)

Physical meaning of the unit impulse signal (Dirac delta function, $\delta(t)$) with the following example:

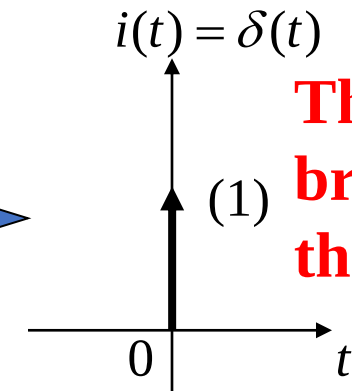


Assume that S, E, and C in the figure are all ideal components (internal resistance = 0). Find current $i(t)$ when S is closed at $t = 0$.

$$i(t) = C \frac{dv_c(t)}{dt}$$



$\tau \rightarrow 0$



The "1" in the bracket represents the strength.

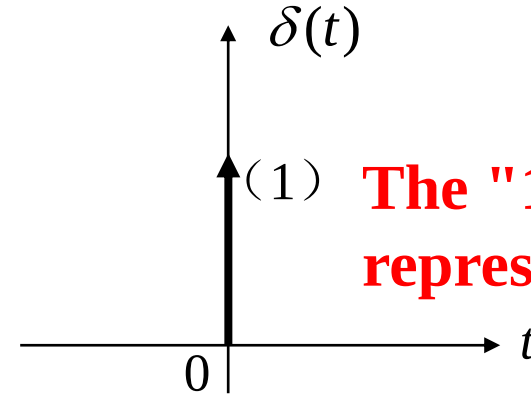
1.4 Singular Signals



Definition of $\delta(t)$

1) By Expression

$$\begin{cases} \delta(t) = 0 & (t \neq 0) \\ \int_{-\infty}^{\infty} \delta(t) dt = 1 \end{cases}$$



The "1" in the bracket represents the strength.

2) $\delta(t)$ is an even function, i.e., $\delta(t) = \delta(-t)$

$$3) \int_{-\infty}^t \delta(\tau) d\tau = \begin{cases} 0 & t < 0 \\ 1 & t > 0 \end{cases} = u(t)$$

$$\int_{-\infty}^t \delta(\tau - t_0) d\tau = u(t - t_0)$$

1.4 Singular Signals



Properties of the Unit Impulse Signal

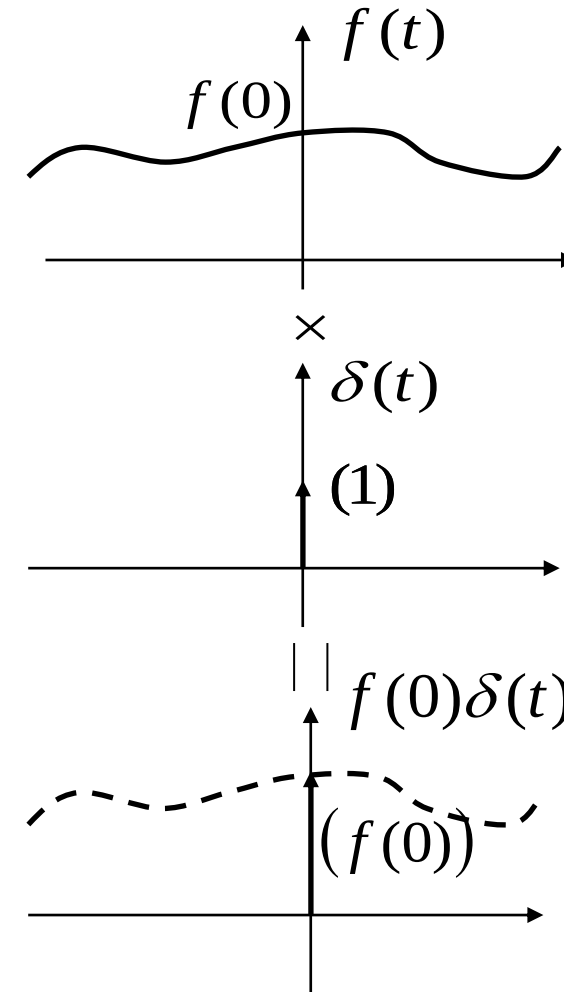
1) Sampling

$$f(t)\delta(t) = f(0)\delta(t) \quad (1)$$

$$\int_{-\infty}^{\infty} \delta(t)f(t)dt = f(0)\int_{-\infty}^{\infty} \delta(t)dt = f(0) \quad (2)$$

$$f(t)\delta(t-t_0) = f(t_0)\delta(t-t_0) \quad (3)$$

$$\int_{-\infty}^{\infty} \delta(t-t_0)f(t)dt = f(t_0) \quad (4)$$



Combining formulas (2) and (4), we can draw the following conclusion:

The impulse function can extract (sample) the function value at the position where the impulse is located.

1.4 Singular Signals



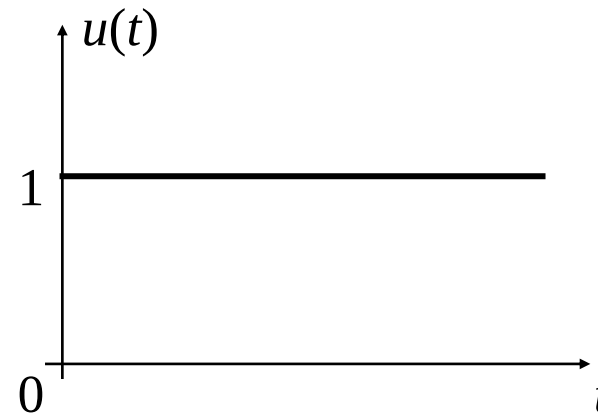
Relationship between $u(t)$ and $\delta(t)$

$$\int_{-\infty}^t \delta(\tau) d\tau = u(t)$$

$$\frac{d}{dt} u(t) = \delta(t)$$

$$\int_{-\infty}^t \delta(\tau - t_0) d\tau = u(t - t_0)$$

$$\frac{d}{dt} u(t - t_0) = \delta(t - t_0)$$



$$f(t) = \frac{d}{dt} [e^{-t} \delta(t)] = (\quad)$$

- ☐ A $\delta(t)$
- ☒ B $\delta'(t)$
- ☐ C $e^{-t} \delta'(t)$
- ☐ D $-e^{-t} \delta'(t)$

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Contents of This Lecture

1.5 Decomposition of Signals

1.6 System Models and Classification

1.7 Linear Time-Invariant Systems

1.8 System Analysis Methods

Learning Objectives of This Lecture

1. Master the method of decomposing any signal into pulse components;
2. Understand different classification methods of systems;
3. Be able to accurately judge the **linearity, time-invariance, and causality** of a system;
4. Initially understand the analysis methods of systems.

1.5.1 Decomposition of Any Signal into the Sum of Even and Odd Components

Even component: $f_e(t) = f_e(-t)$

Odd component: $f_o(t) = -f_o(-t)$

Any signal can be decomposed into the sum of an even component and an odd component, i.e.,

$$f(t) = f_e(t) + f_o(t) \quad (1)$$

$$f(-t) = f_e(t) - f_o(t) \quad (2)$$

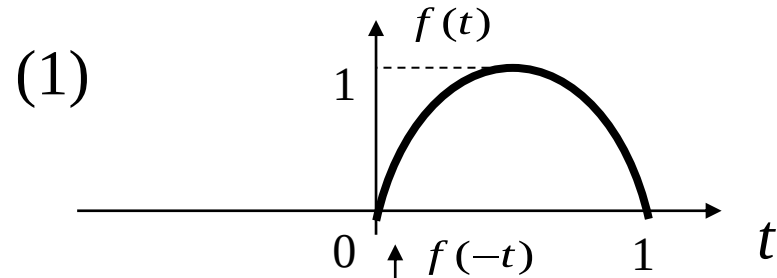
$$(1) + (2): \quad f_e(t) = \frac{1}{2}[f(t) + f(-t)]$$

$$(1) - (2): \quad f_o(t) = \frac{1}{2}[f(t) - f(-t)]$$

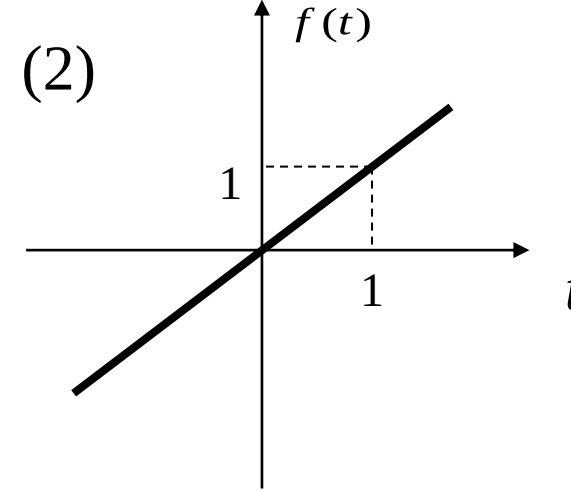
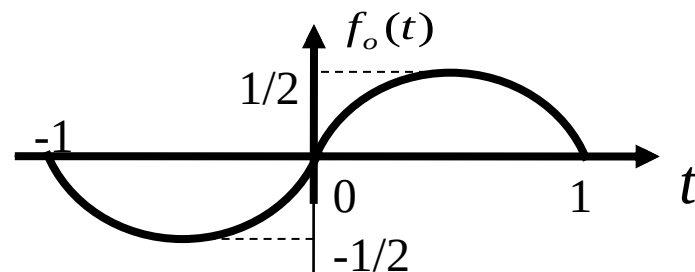
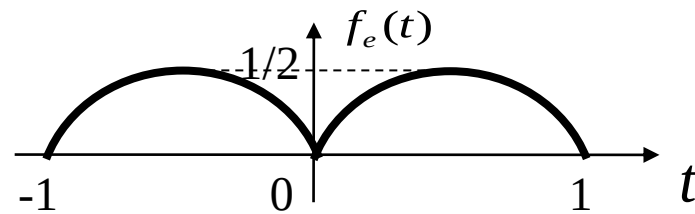
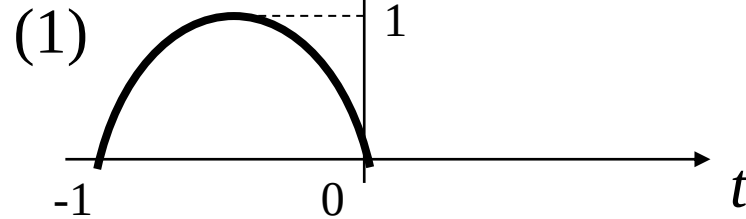
1.5 Decomposition of Signals



Example 1-9: Find the even and odd components of the signals shown in the figures.



Solution:



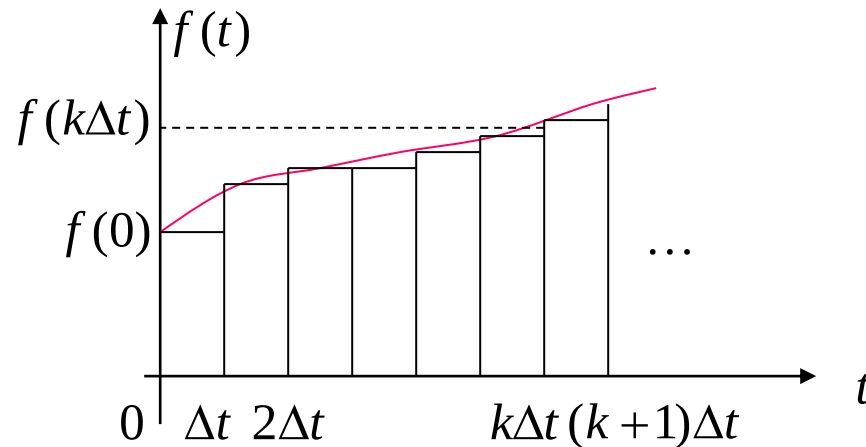
(2) This is an odd signal, so

$$f_o(t) = f(t)$$

$$f_e(t) = 0$$

1.5.2 Decomposition of Any Signal into Pulse Components

A signal can be approximately decomposed into the sum of many pulse components. A common decomposition is the decomposition into **rectangular narrow pulse components**, and the limit of the combination of narrow pulses is the **superposition of impulse signals**.

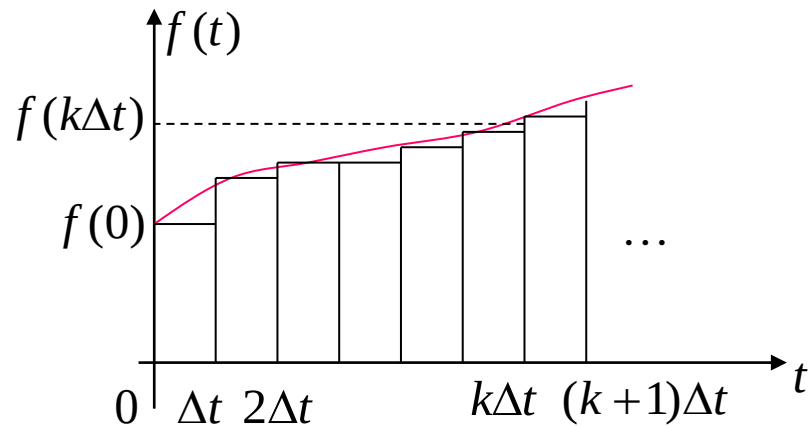


When $t = 0$, the first rectangular pulse is

$$f(0)[u(t) - u(t - \Delta t)]$$

$$\lim_{\Delta t \rightarrow 0} \frac{f(0)[u(t) - u(t - \Delta t)]}{\Delta t} \Delta t = \lim_{\Delta t \rightarrow 0} f(0)\delta(t)\Delta t$$

1.5 Decomposition of Signals

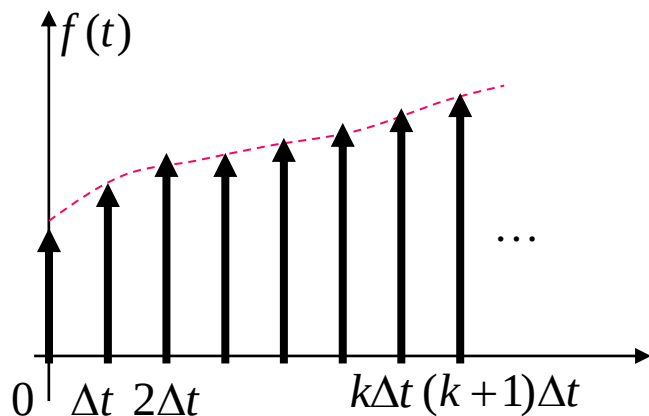


When $t = k\Delta t$, the $(k+1)$ -th rectangular pulse is

$$f(k\Delta t)\{u(t - k\Delta t) - u[t - (k + 1)\Delta t]\}$$

$$\lim_{\Delta t \rightarrow 0} f(k\Delta t) \frac{\{u(t - k\Delta t) - u[t - (k + 1)\Delta t]\}}{\Delta t} \Delta t$$

$$= \lim_{\Delta t \rightarrow 0} f(k\Delta t) \delta(t - k\Delta t) \Delta t$$



Superimposing the 0 to n rectangular pulses mentioned above, we get the expression of $f(t)$, i.e.,

$$f(t) = \lim_{\Delta t \rightarrow 0} \sum_{k=0}^n f(k\Delta t) \delta(t - k\Delta t) \Delta t$$

$$\text{When } \Delta t \rightarrow 0, \Delta t \rightarrow d\tau, k\Delta t \rightarrow \tau, \lim_{\Delta t \rightarrow 0} \sum_{k=0}^n \rightarrow \int_{0^-}^t$$

$$f(t) = \int_{0^-}^t f(\tau) \delta(t - \tau) d\tau$$

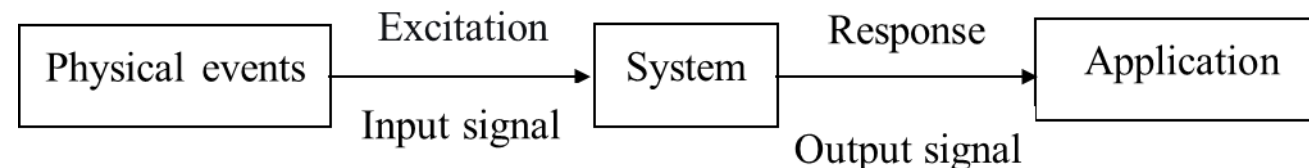
1.5.3 Decomposition of a Signal into Orthogonal Function Components

If a signal is represented by an orthogonal function set, then each component composing the signal is mutually orthogonal.

For example, the trigonometric function set composed of sine and cosine signals of various harmonics is an orthogonal function set. A periodic signal $f(t)$ can be represented by the linear combination of these trigonometric functions as long as it satisfies the Dirichlet conditions, which is called the trigonometric form of the Fourier series of $f(t)$. Similarly, $f(t)$ can also be expanded into the exponential form of the Fourier series. (Content of Chapter 3)

Definition of a System

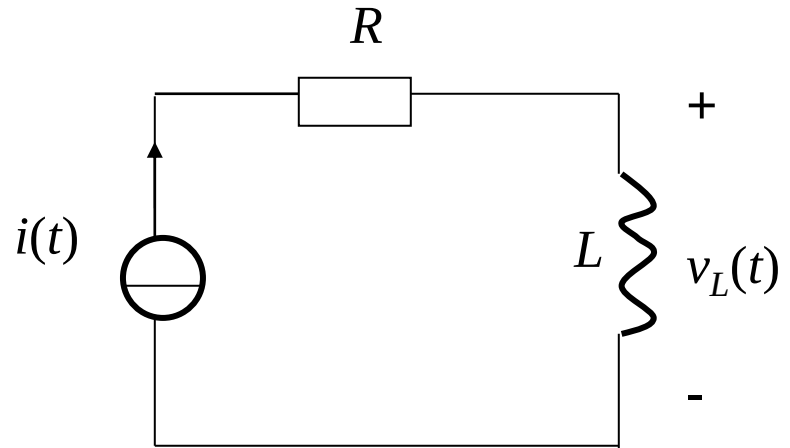
- A system is an integrated whole composed of several interrelated and interacting things, with certain or specific functions. Examples include communication systems and radar systems.
- The concept of a system is not only applicable to various fields of natural sciences but also to social sciences, such as political structures and economic organizations.
- A common feature of systems in many different fields is that all systems respond to the signals applied to them (i.e., the **input signals** of the system, also called **excitations**) and generate other signals (i.e., the **output signals** of the system, also called **responses**). The function of a system is reflected in what kind of output signal is generated by a certain input signal.
- Each system has its own **mathematical model**.



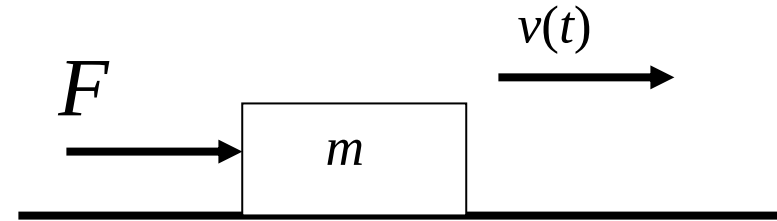
1.6 System Models and Classification



1.6.1 Mathematical Models of Systems



$$v_L(t) = L \frac{di(t)}{dt}$$



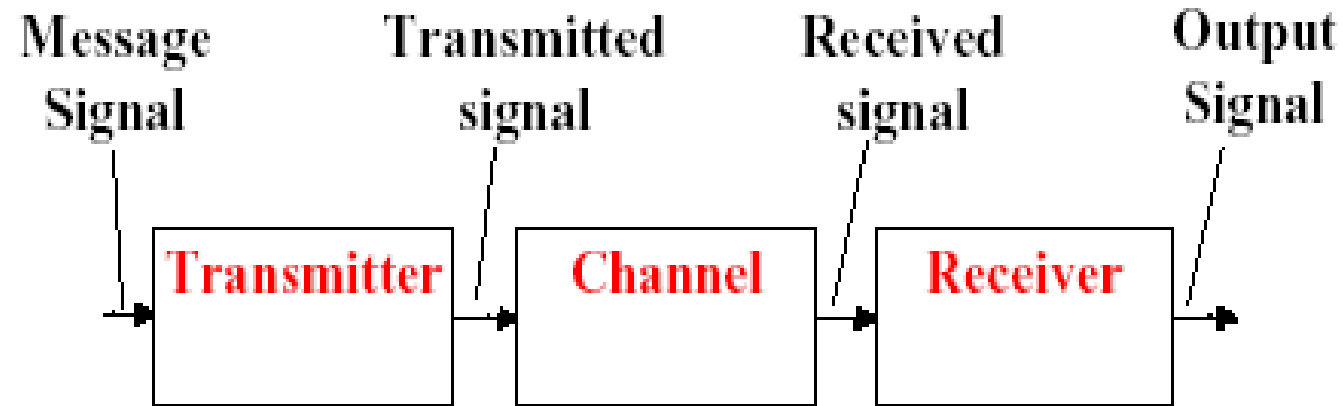
$$F = ma = m \frac{dv(t)}{dt}$$

Two different systems may have the same mathematical model, and even physical systems and non-physical systems may have the same mathematical model. Systems with the same mathematical model are called similar systems.

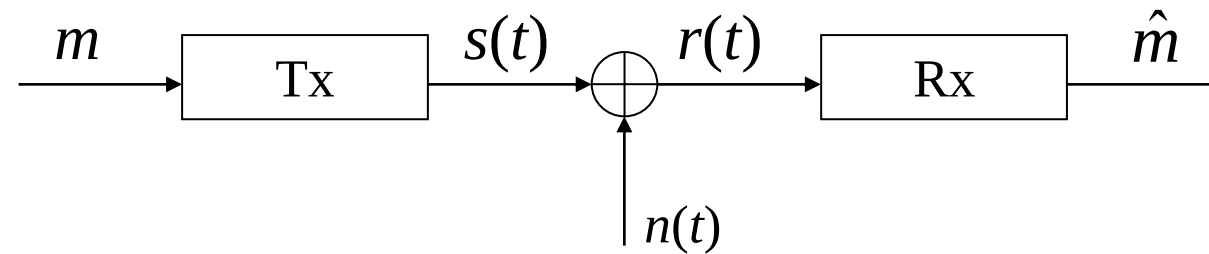
1.6 System Models and Classification



Communication System

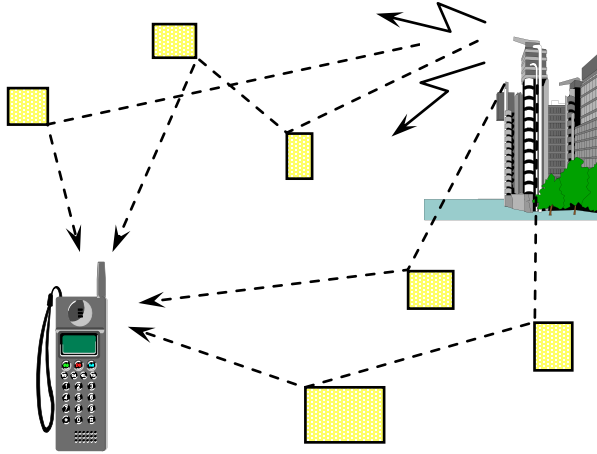


Additive White Gaussian Noise (AWGN) Channel Model

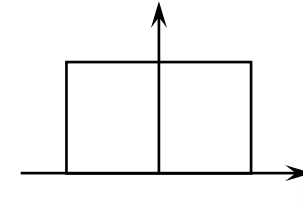


$$r(t) = s(t) + n(t)$$

Multipath Channel Model

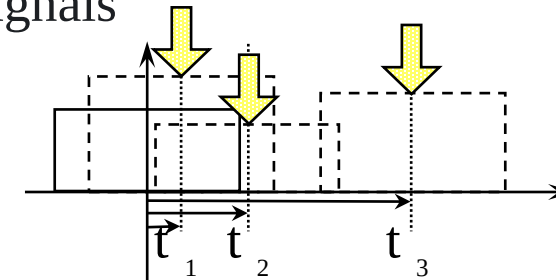


Transmitted signal



Received signals

Inter-symbol interference



1.6.2 Classification of Systems

(1) Continuous-Time Systems and Discrete-Time Systems

- The mathematical model of a continuous-time system is a differential equation.
- The mathematical model of a discrete-time system is a difference equation.

(2) Instantaneous Systems (Memoryless Systems) and Dynamic Systems (Memory Systems)

- The mathematical model of an instantaneous system is an algebraic equation, such as a resistor circuit.
- The mathematical model of a dynamic system is a differential equation or a difference equation, such as an RC or RL circuit.

(3) Lumped Parameter Systems and Distributed Parameter Systems

- The mathematical model of a lumped parameter system is an ordinary differential equation.
- The mathematical model of a distributed parameter system is a partial differential equation.

(4) Linear Systems and Nonlinear Systems

- A system with superposition and homogeneity (also called scalability) is called a linear system.
- A system that does not satisfy superposition or homogeneity is called a nonlinear system.

(5) Time-Varying Systems and Time-Invariant Systems

- Time-varying system: The parameters of the system change with time.
- Time-invariant system: The parameters of the system do not change with time.

(6) Single-Input Single-Output (SISO) Systems and Multiple-Input Multiple-Output (MIMO) Systems

- SISO system: Only accepts one excitation and generates one response.
- MIMO system: The system has more than one excitation signal and more than one response signal. For example, the massive MIMO system in 5G.

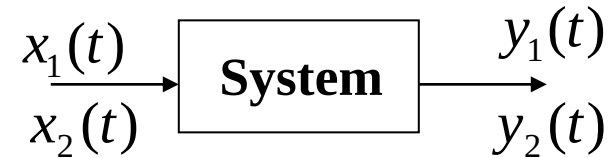
(7) Invertible Systems and Non-Invertible Systems

- Invertible system: Different excitations generate different responses.
 - Non-invertible system: Different excitations generate the same response.
 - For each invertible system, there exists an “inverse system”. When the original system and its inverse system are cascaded, the output signal is the same as the input signal.
 - Examples:
 - An invertible system: $r(t) = 3e(t)$
Its inverse system: $r(t) = e(t) / 3$
 - A non-invertible system: $r(t) = e^2(t)$
- When the excitations are $e(t) = 1$ and $e(t) = -1$, the response $r(t)$ is 1 in both cases. That is, different excitations generate the same response, so it is a non-invertible system.

1.7.1 Linearity

Linearity includes superposition and homogeneity (also called scalability).

1. Superposition



If $x(t) = x_1(t) + x_2(t)$, $y(t) = y_1(t) + y_2(t)$,
the system is said to satisfy superposition.

2. Homogeneity

If $x(t) = ax_i(t)$, $y(t) = ay_i(t)$,

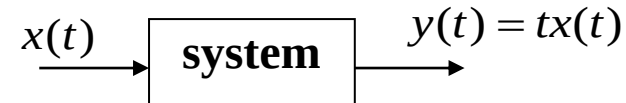
the system is said to satisfy homogeneity.

A system that satisfies both superposition and homogeneity is called a **linear system**.

1.7 Linear Time-Invariant Systems



Example 1-10: Assume the input-output relationship of a system is $y(t) = tx(t)$. Determine whether the system is linear.



Solution:

Check superposition:

$$y_1(t) = tx_1(t) \quad y_2(t) = tx_2(t) \quad x(t) = x_1(t) + x_2(t)$$

$$y(t) = t[x_1(t) + x_2(t)] = tx_1(t) + tx_2(t) = y_1(t) + y_2(t)$$

Superposition is satisfied.

Check homogeneity:

$$x(t) = ax_1(t) \quad y(t) = tx(t) = atx_1(t) = ay_1(t)$$

Homogeneity is satisfied.

$$x(t) = \sum_{k=1}^N a_k x_k(t) \quad y(t) = \sum_{k=1}^N a_k y_k(t)$$

Hence, the system is **linear**.

Assume the input-output relationship of a system is $y(t) = ax(t) + b$ ($b \neq 0$) ,
Determine whether the system is linear.

☐ A Yes

☒ B No

提交

1.7 Linear Time-Invariant Systems



Example 1-11: Assume the input-output relationship of a system is $y(t) = ax(t) + b$ ($b \neq 0$), Determine whether the system is linear.

Solution:

Check superposition:

$$y_1(t) = ax_1(t) + b \quad y_2(t) = ax_2(t) + b$$

$$x(t) = x_1(t) + x_2(t)$$

$$y(t) = a[x_1(t) + x_2(t)] + b \neq y_1(t) + y_2(t)$$

Therefore, the system does not satisfy superposition.

Check homogeneity:

$$x(t) = cx_1(t) \quad y(t) = ax(t) + b = acx_1(t) + b \neq cy_1(t)$$

Therefore, the system does not satisfy homogeneity.

The system is **not linear**.

1.7 Linear Time-Invariant Systems



For a linear system, if the input of the system is zero at all times, the output must be zero, i.e., zero input produces zero output.

$$x(t) = \sum_{k=1}^N a_k x_k(t) = \sum_{k=1}^N 0 \cdot a_k = 0$$

$$y(t) = \sum_{k=1}^N a_k y_k(t) = \sum_{k=1}^N 0 \cdot a_k = 0$$

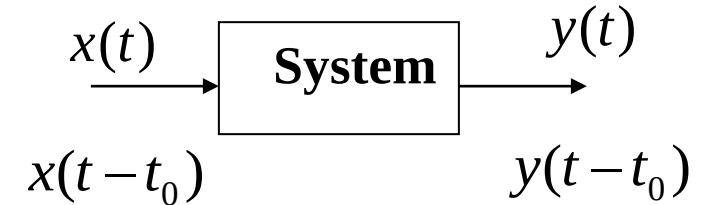
However, for the system,

$$y(t) = ax(t) + b = a \cdot 0 + b = b$$

when the input $x(t) = 0$, the output $y(t) = b \neq 0$. This non-zero output is called the **zero-input response** of the system.

1.7.2 Time-Invariance

Time-invariant system: The parameters of the system do not change with time; if the excitation is delayed, the response is delayed by the same amount.



Example 1-12: Determine whether the systems satisfying the following input-output relationships are time-varying: (1) $y(t) = tx(t)$, (2) $y(t) = ax(t) + b$.

Solution: (1) $y_1(t) = tx(t - t_0) \neq y(t - t_0)$

$$y(t - t_0) = (t - t_0)x(t - t_0)$$

However, $y(t - t_0) = (t - t_0)x(t - t_0)$. Since $y_1(t) \neq y(t - t_0)$, the system is **time-varying**.

$$(2) \quad y_1(t) = ax(t - t_0) + b = y(t - t_0)$$

So the system is **time-invariant**.

Assume the input-output relationship of a system is $y(t) = x(2t)$. Determine whether it is a time-invariant system.

- ☐ A Time-Invariant
- ☒ B Time-Varying

提交

1.7 Linear Time-Invariant Systems



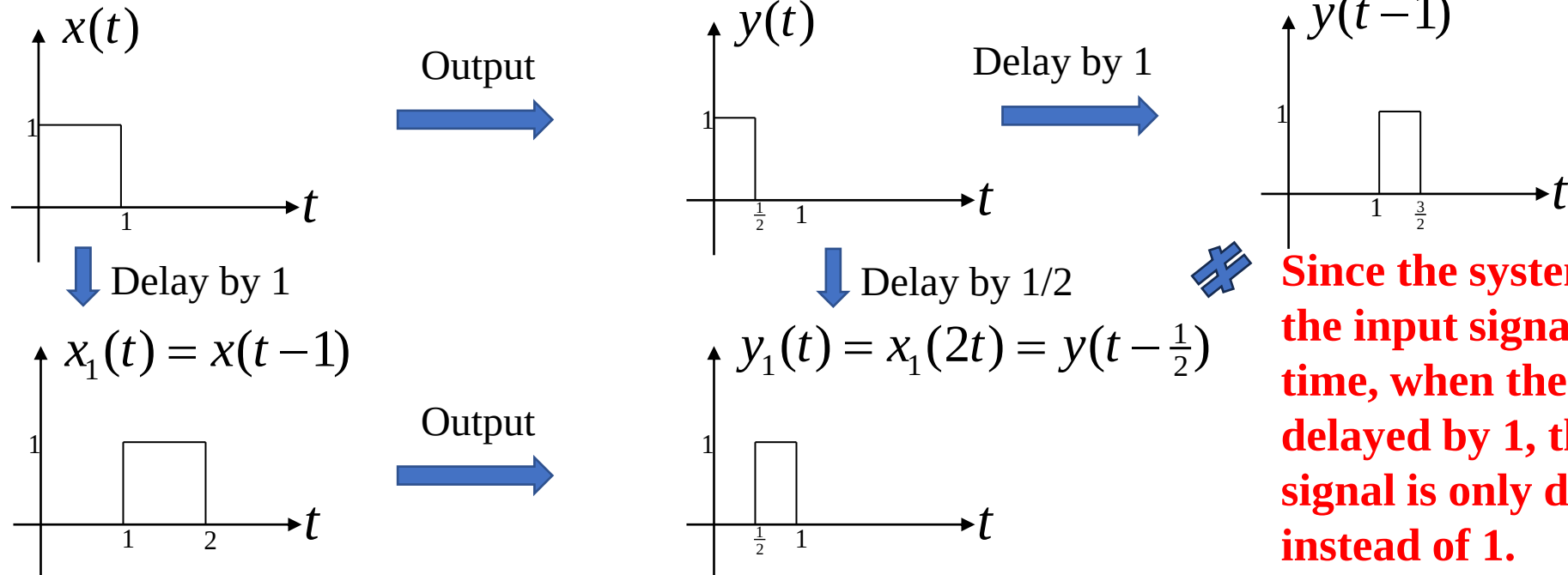
To determine whether a system satisfies a certain property, **we only need to find one example that does not satisfy it** to prove that the system does not have that property.

Example 1-13: Assume the input-output relationship of a system is $y(t) = x(2t)$.

Determine whether it is a time-invariant system.

Solution: Define the delayed input of $x(t)$ by t_0 as $x_1(t) = x(t - t_0)$, and the corresponding output is $y_1(t) = x_1(2t) = x(2t - t_0)$. However, the direct delay of $y(t)$ by t_0 gives $y(t - t_0) = x[2(t - t_0)] = x(2t - 2t_0)$. The two are not equal, so the system is **time-varying**.

E.g.,



Since the system compresses the input signal by 2 times in time, when the input signal is delayed by 1, the output signal is only delayed by 1/2 instead of 1.

Example 1-14: Determine whether $y(t) = x(-t)$ is a time-invariant system.

Solution: $y(t) = x(-t)$

$$x_1(t) = x(t - t_0)$$

$$y_1(t) = x(-t - t_0)$$

$$y(t - t_0) = x(-t + t_0) \neq y_1(t)$$

This system only reverses the excitation, i.e., multiplies t in x by -1 .

Since $y_1(t) \neq y(t - t_0)$, the system is **time-varying**.

Intuitive Judgment Method for Time-Invariance:

If there is a time-varying coefficient in front of $x(\cdot)$ (e.g., $tx(t)$), or reverse (e.g., $x(-t)$), or scaling transformation (e.g., $x(2t)$), then the system is time-varying.

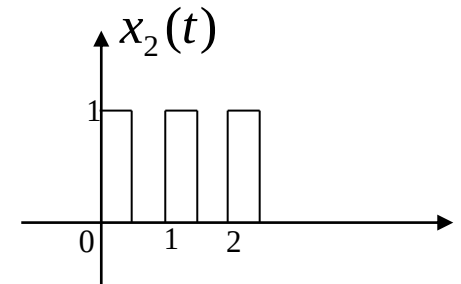
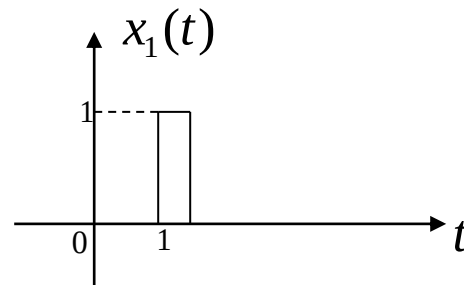
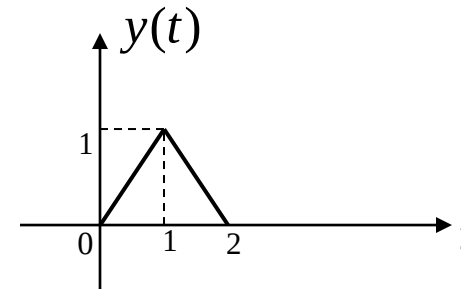
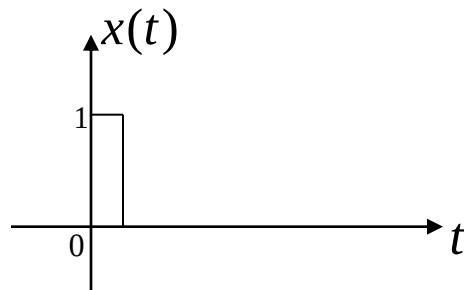
1.7 Linear Time-Invariant Systems



A system that satisfies both linearity and time-invariance is called a **Linear Time-Invariant (LTI)** system, which can be expressed as:

$$x(t) = \sum_{k=1}^N a_k x_k(t - t_k) \quad y(t) = \sum_{k=1}^N a_k y_k(t - t_k)$$

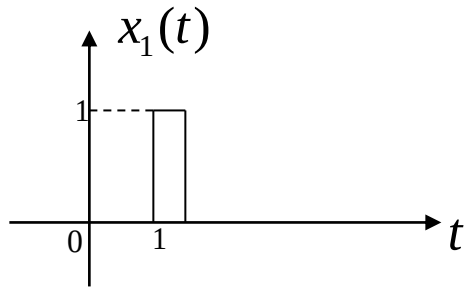
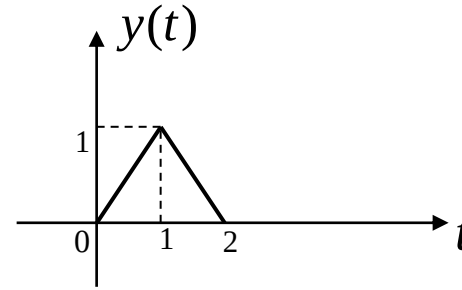
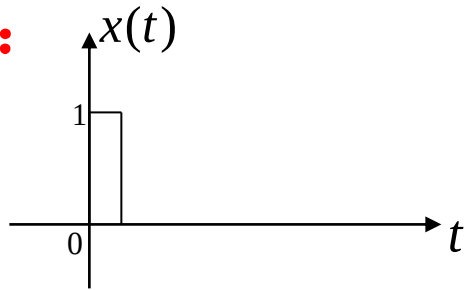
Example 1-15: Assume the input-output relationship of an LTI system is described in the figure. Draw the waveforms of outputs $y_1(t)$ and $y_2(t)$ when the inputs are $x_1(t)$ and $x_2(t)$ respectively.



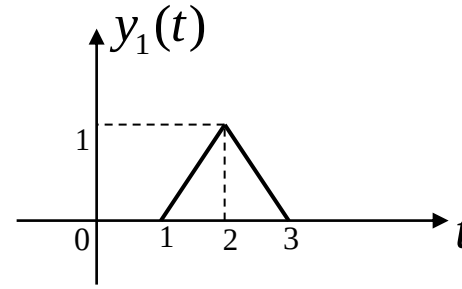
1.7 Linear Time-Invariant Systems



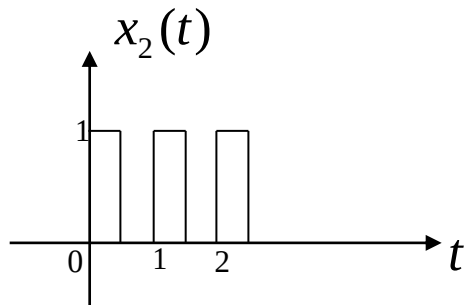
Solution:



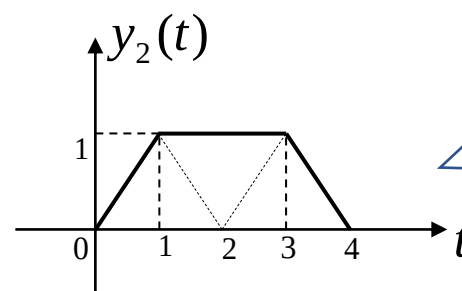
$$x_1(t) = x(t-1)$$



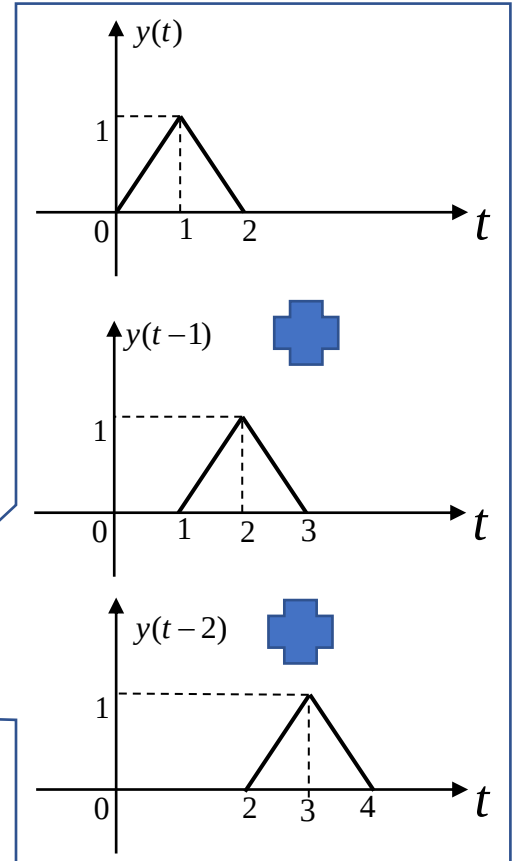
$$y_1(t) = y(t-1)$$



$$x_2(t) = x(t) + x(t-1) + x(t-2)$$

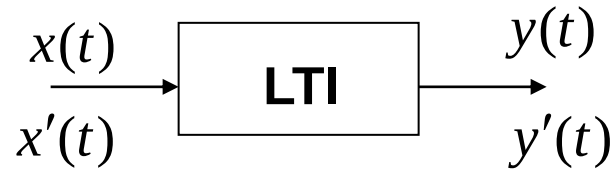


$$y_2(t) = y(t) + y(t-1) + y(t-2)$$

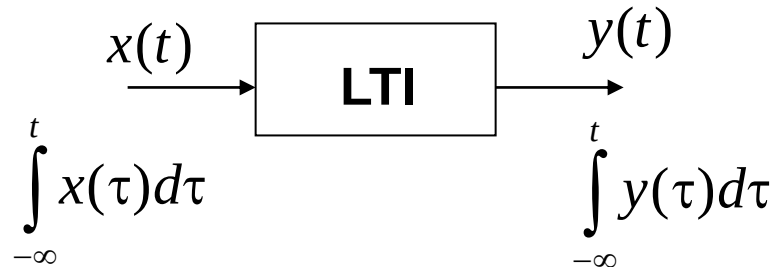


1.7.3 Differentiation and Integration Properties of Continuous-Time Systems

1. Differentiation Property



2. Integration Property



1.7.4 Causality

A causal system is a system where the response at time t_0 depends only on the input at time $t \leq t_0$. Otherwise, it is a non-causal system.

Examples:

Causal system: $r(t) = e(t - 1)$ (delay system)

Non-causal system: $r(t) = e(t + 1)$ (advance system)

(The response at time $t = 0$ is determined by the excitation at time $t = 1$, so it is a non-causal system.)

Non-causal system: $r(t) = e(2t)$ (time-compressed system)

Non-causal systems are common in fields such as speech signal processing, meteorology, stock market analysis, and demography.

1.8 Analysis Methods of LTI Systems



Continuous –Time



Discrete –Time



Time-Domain Analysis

Transform-Domain Analysis

Frequency-Domain
Complex Frequency-Domain

State-Space Analysis

Engineering Applications

Classification by Mathematical Description

- Input-Output Analysis:** An n -th order differential (difference) equation, suitable for SISO systems (Chapters 2, 3, 4, 6 and 7)
- State Variable Analysis:** A system of n first-order differential equations, suitable for MIMO and nonlinear systems (Chapter 9)

Classification by Solving Methods of System Mathematical Models

- Time-Domain Analysis:** Solve the response directly in the time domain without any transformation (Chapters 2 and 6)
- Transform-Domain Analysis:** Transform the time functions of signals and system models into functions in the corresponding transform domain, such as Fourier Transform (Chapters 3 and 5), Laplace Transform (Chapter 4), Z-Transform (Chapter 7), etc.

- Homework: 1.64 (iii)(iv)(v)(a)(d)(f)(i)(k) in the textbook.
- Textbook: Simon Haykin and Barry Van Veen, Signals and Systems (Second Edition), John Wiley & Sons, 2002, 9780471164746.
- You do not need to submit the homework, but you are highly recommended to do the exercises. Refer to the solutions after you finish the homework.